

Optimal Policy:

Def (Optimal Policy):

$$\exists \pi^*, \forall s \in S, \text{ for } \forall \pi \neq \pi^*, \quad V_{\pi^*}(s) \geq V_{\pi}(s)$$

Bellman Optimality Equation:

Thy. (BOE)

$$V(s) = \max_{\pi} \sum_a \pi(a|s) \left(\sum_r p(r|s,a) r + \gamma \sum_{s'} p(s'|s,a) V(s') \right), \quad \forall s \in S$$

$$V = \max_{\pi} (r_{\pi} + \gamma P_{\pi} V)$$

Solve:

$$V(s) = \max_{\pi} \sum_a \pi(a|s) \underbrace{g(s,a)}_{\text{known}}$$

let

$$\Rightarrow \textcircled{1} \pi(a|s) = \begin{cases} 1, & a = a^* \\ 0, & a \neq a^* \end{cases}$$

$$\text{where } a^* = \arg \max_a g(s,a)$$

$$\textcircled{2} V(s) = \max_{\pi} \sum_a \pi(a|s) g(s,a) = \max_{a \in A(s)} g(s,a).$$

$$\text{let } f(v) = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v)$$

then BOE is: $v = f(v)$

$$\text{where } [f(v)]_s = V(s) = \max_{\pi} \sum_a \pi(a|s) g(s,a)$$

$f(v)$ is a **contraction mapping**: $\|f(v_1) - f(v_2)\| \leq \gamma \|v_1 - v_2\|$

therefore, according to **contraction mapping theorem**:

iteratively:

$$v_{k+1} = f(v_k) = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v_k)$$

$$\{v_k\} \longrightarrow \underline{v^*} \text{ (exponentially fast)}$$

exist & is unique & is optimal

$$\Rightarrow \begin{cases} \pi^* = \arg \max_{\pi} (r_{\pi} + \delta P_{\pi} v^*) \\ v^* = r_{\pi^*} + \delta P_{\pi^*} v^* \end{cases}$$